Yuxiang Lin

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I am interested in the intersection of computer science and discrete mathematics, particularly in combinatorial and graph algorithms, as well as program analysis. I am drawn to both the theoretical properties of combinatorial objects and the practical work of reasoning about program behavior. My background in algorithmic programming contests and large-scale LLM systems reflects both my analytical ability and engineering experience.

Education

Duke Kunshan University (DKU) & Duke University Dual Degree, Class of 2026

Duke Kunshan University **Kunshan, China**

B.S. in Applied Math & Computational Sciences; Computer Science

- GPA: 3.962/4.0; Dean's List with Distinction (top 10%), 4 semesters (Fall 2022–Spring 2024)
- Scholarships: Hong Scholarship (admission), 2023 National Encouragement Scholarship (9 recipients in DKU)
- Courses: MATH 305 (Linear Algebra, A), MATH 307 (Complex Analysis, A), COMPSCI 308 (Algorithms, A+)

Duke University **Durham, U.S.**

B.S. in Interdisciplinary Studies; Applied Math & Computational Sciences; Computer Science

- Courses: COMPSCI 553 (Compiler Construction, A+), COMPSCI 650 (Parallel Computer Architecture)

Publications

- Shao, J., **Lin, Y.**, Lohani, M. P., Miao, Y., and Luo, B., "Do LLM Agents Know How to Ground, Recover, and Assess? A Benchmark for Epistemic Competence in Information-Seeking Agents", arXiv e-prints, arXiv:2509.22391, 2025.

Research Experience

Prof. Danfeng Zhang (Duke CS+ Summer Program) **Durham, U.S.**

Researcher in Program Analysis for Side-Channel Attacks *05/2025–Present*

- Modifying the source code of *KLEE* to detect control-flow and memory-access side-channel vulnerabilities
- Implemented product-program construction in *KLEE* to enable control-flow side-channel detection
- Developed an automated benchmarking suite to verify *KLEE*-identified positives and generate reports
- Discovered two previously human-verified vulnerabilities in peer-reviewed work to be false positives
- Revealed control-flow dependencies among vulnerabilities reported in prior studies

Prof. Italo Simonelli (Undergraduate Signature Work) **Kunshan, China**

Researcher in Matroids and Sunflower Lemma *01/2024–Present*

- Proposing "weak matroids," a new combinatorial structure obtained by relaxing the matroid axioms
- Exploring connections of weak matroids to digraphs, the sunflower lemma, and other structures
- Formulated and analyzed a variant of the sunflower lemma
- Using C++ and the Boost Graph Library for combinatorial enumeration experiments

DKU Edge Intelligence Lab (Prof. Bing Luo, DKU Summer Research Scholars) **Kunshan, China**

Lead Researcher and Developer of ChatDKU *05/2024–01/2025*

- Led the development of *ChatDKU*, an AI assistant for the DKU community, contributing over 26,000 lines of code
- Implemented an agentic retrieval-augmented generation (RAG) architecture with LlamaIndex and DSPy
- Built scalable inference infrastructure using SGLang, vLLM, Text Embeddings Inference, Flask, and Redis
- Contributed upstream fixes to open-source projects LlamaIndex and DSPy

Academic Contest

DKU Programming Contest Club **Kunshan, China**

Member of DKU ICPC Teams *10/2022–11/2024*

- Silver Medal (Rank 108/368) at the 2023 ICPC Asia Xi'an Regional Contest
- Bronze Medal (Rank 149/281) at the 2022 ICPC Asia-East Continent Final Contest
- Silver Medal (Rank 100/417) at the 2022 ICPC Asia Nanjing Regional Contest
- Gold Medal (Rank 20/452) at the 2022 ICPC Asia Hefei Regional Contest

Projects & Presentations

DKU Innovation Incubator **Kunshan, China**
Founder and Leader of the Affordable Embroidery Machine Project *09/2023–05/2025*
- Received Certificate of Completion for College Student Innovation and Entrepreneurship Training Program
- Designed an affordable and scalable CNC embroidery machine using FreeCAD
- Constructed a 3D-printed prototype and implemented motion control using Arduino

DKU Edge Intelligence Lab (Prof. Bing Luo) **Kunshan, China**
Backend Developer of FedCampus *01/2024–04/2024*
- Maintained the Django backend for a federated learning and analytics platform for DKU with 90 active users

DKU Humanities Research Center **Kunshan, China**
Undergraduate Humanities Research Conference Presenter *04/2024*
- Presented paper "Hacktivism and the Reconfiguration of Rights and Power Relations in the Digital World"

Teaching & Mentoring

Prof. Jiang Long **Kunshan, China**
COMPSCI 201 (Introduction to Programming and Data Structures) Teaching Assistant *08/2023–12/2023*
- Provided in-depth, beyond-curriculum support to 20 students during weekly office hours

DKU Academic Resource Center **Kunshan, China**
STATS 102 (Introduction to Data Science) Peer Tutor *08/2023–05/2024*
- Trained for 11 hours in Tutor Essentials, a CRLA-aligned tutor training course
- Provided tutoring sessions twice a week; proficient in matplotlib and scikit-learn

Leadership & Service

DKU Programming Contest Club **Kunshan, China**
President, Vice President, and Marketing Coordinator *10/2022–12/2024*
- The club received 2023 DKU Academic Student Organization of the Year Award
- I received 2023 DKU Student Leader of the Year Award and 2023 DKU Community Service Award
- Co-founded DKU Programming Contest Club with 100 members and hosted more than 30 seminars on algorithms
- Led the club members to win 7 medals in ICPC, including 3 gold medals

DKU Metal Club **Kunshan, China**
Secretary and Vice President *10/2022–12/2024*
- Co-founded DKU Metal Club, led recruitment for 160 members
- Coordinated the Iridodonesis Debut metal music show with 40 attendees

DKU Career Services Office **Kunshan, China**
Career Fair Volunteer *03/2024*
- Assisted in organizing a career fair with 55 companies participating

DKU Chinese Enrollment Management **Xiamen, China**
Student Ambassador *10/2023–03/2024*
- Received Third Prize and On-site Most Popular Award
- Presented DKU to more than 400 students from two high schools in Xiamen

DKU First Year Experience **Kunshan, China**
Orientation Leader *08/2023–10/2023*
- Volunteered in the orientation week for over 600 first-year students
- Teaching assistant for DKU 101, a course for all 600 first-year students

Additional Information

Languages: Chinese Mandarin (Native), English (Proficient)

Skills: C++ (used extensively over nine years of competitive programming), Python (used extensively for ChatDKU and other projects), Standard ML (used for compiler construction), Java, LaTeX, SQL, GNU/Linux, Git, OpenGL, CAD modeling

Interests: Heavy metal music, guitar, motorcycling, bicycling, basketball, swimming, video games